13 The density of ethanol is 1.09 g/cm^3
The density of propylene is 0.97 g/cm^3

60 litres of ethanol are mixed with 128 litres of propylene to make 188 litres of antifreeze.

Work out the density of the antifreeze.
Give your answer correct to 2 decimal places.

Ethanol: 60 000 × 1.09 = 65 400 g.

1 mark
Find ethanol mass.

Propylene: 128 000 × 0.97 = 124 160 g.

1 mark
Find propylene mass.

Total mass = 189 560 g.

Total volume = 188 000 cm^3.

1 mark
Total mass ÷ total volume.

Density = 189 560 ÷ 188 000 = 1.00829...

= 1.01 g/cm^3 (2 decimal places).

1 mark
1.01 g/cm^3 to 2 d.p.

1.01 ..... g/cm^3

(Total for Question 13 is 4 marks)